

MATLAB Simulation Codes for [1, Sect. IV]

These MATLAB codes reproduce the simulation results from [2, Sect. 6]

CACC for Heterogeneous Platoons With Actuation And Communication Delays

a) Vehicle dynamics: We consider a heterogeneous string of vehicles each one modeled by the following third-order, linear system with actuator delay that describes vehicle dynamics

$$\dot{s}_i(t) = v_{i-1}(t) - v_i(t), \quad (1)$$

$$\dot{v}_i(t) = a_i(t), \quad (2)$$

$$\dot{a}_i(t) = -\frac{1}{\tau_i}a_i(t) + \frac{1}{\tau_i}u_i(t-D), \quad (3)$$

$i = 1, \dots, N$, where $s_i = x_{i-1} - x_i - l_i$ and x_i is the position of vehicle i and l_i is its length, v_i is vehicle speed, a_i is vehicle acceleration, τ_i is lag, capturing, engine dynamics, u_i is the individual vehicle's control variable, $D \geq 0$ is input delay, and $t \geq 0$ is time.

b) Nominal control design: Without input and communication delays, the following MPF topology-based control strategy is constructed

$$\begin{aligned} u_i(t) = & \tau_i \alpha_i \sum_{n_i=1}^{m_i} (m_i - n_i + 1) \frac{h_{i-n_i+1}}{h_i} \left(\frac{s_{i-n_i+1}(t)}{h_{i-n_i+1}} - v_{i-n_i+1}(t) \right) + \tau_i b_i \left(\sum_{n_i=1}^{m_i} v_{i-n_i}(t) - m_i v_i(t) \right) \\ & + \tau_i c_i \left(\sum_{n_i=1}^{m_i} a_{i-n_i}(t) - m_i a_i(t) \right), \end{aligned} \quad (4)$$

for $i \geq m_i$, where $\alpha_i > 0$, $b_i > 0$, and $c_i > 0$ are design parameters, and $h_i > 0$ is a desired time-headway. Controller (4) aims to regulate v_i to the average speed of the preceding vehicles with which the ego vehicle communicates. While the first term in (4) indicates the objective of regulating the spacing of each vehicle to $h_i v_i$, as customary in constant time-headway (CTH)-based strategies.

Predictor-Feedback CACC Design

The predictor-feedback control laws for system (1)–(3) are given by

$$u_i(t) = K_i^T q_i(t), \quad (5)$$

where

$$q_i(t) = e^{\Gamma_i D} \bar{x}_i(t) + \int_{t-D}^t e^{\Gamma_i(t-\theta)} B_i u_i(\theta) d\theta + \sum_{j=1}^{m_i} \int_{t-D}^t e^{\Gamma_i(t-\theta)} B_{i-j} u_{i-j,m}(\theta) d\theta, \quad (6)$$

with

$$q_i = \begin{bmatrix} q_{i,s_i} \\ q_{i,s_i-1} \\ \vdots \\ q_{i,s_i-m_i+1} \\ q_{i,v_i} \\ q_{i,v_i-1} \\ \vdots \\ q_{i,v_i-m_i} \\ q_{i,a_i} \\ q_{i,a_i-1} \\ \vdots \\ q_{i,a_i-m_i} \end{bmatrix}, \quad \bar{x}_i = \begin{bmatrix} s_i \\ s_{i-1,m} \\ \vdots \\ s_{i-m_i+1,m} \\ v_i \\ v_{i-1,m} \\ \vdots \\ v_{i-m_i,m} \\ a_i \\ a_{i-1,m} \\ \vdots \\ a_{i-m_i,m} \end{bmatrix}, \quad \Gamma_i = \begin{bmatrix} \mathbf{0}_{m_i \times m_i} & M_{i,1_{m_i \times (m_i+1)}} & \mathbf{0}_{m_i \times (m_i+1)} \\ \mathbf{0}_{(m_i+1) \times m_i} & \mathbf{0}_{(m_i+1) \times (m_i+1)} & M_{i,2_{(m_i+1) \times (m_i+1)}} \\ \mathbf{0}_{(m_i+1) \times m_i} & \mathbf{0}_{(m_i+1) \times (m_i+1)} & M_{i,3_{(m_i+1) \times (m_i+1)}} \end{bmatrix}, \quad (7)$$

and

$$M_{i,1_{m_i \times (m_i+1)}} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 \\ & \ddots & \ddots & \ddots & \\ 0 & 0 & \dots & -1 & 1 \end{bmatrix}, \quad M_{i,2_{(m_i+1) \times (m_i+1)}} = \mathbb{I}_{(m_i+1) \times (m_i+1)}, \quad (8)$$

$$M_{i,3_{(m_i+1) \times (m_i+1)}} = \begin{bmatrix} -\frac{1}{\tau_i} & 0 & \dots & 0 \\ & \ddots & \ddots & \\ 0 & \dots & 0 & -\frac{1}{\tau_{i-m_i}} \end{bmatrix}, \quad B_{i-j} = \begin{bmatrix} \mathbf{0}_{2m_i+1+j} \\ \frac{1}{\tau_{i-j}} \\ \mathbf{0}_{m_i-j} \end{bmatrix}, \quad B_i = \begin{bmatrix} \mathbf{0}_{2m_i+1} \\ \frac{1}{\tau_i} \\ \mathbf{0}_{m_i} \end{bmatrix}, \quad (9)$$

$$K_i = \begin{bmatrix} m_i \frac{\tau_i \alpha_i}{h_i} \\ (m_i - 1) \frac{\tau_i \alpha_i}{h_i} \\ \vdots \\ \frac{\tau_i \alpha_i}{h_i} \\ -m_i \tau_i (\alpha_i + b_i) \\ \tau_i b_i - \tau_i \alpha_i (m_i - 1) \frac{h_{i-1}}{h_i} \\ \vdots \\ \tau_i b_i \\ -m_i \tau_i c_i \\ \tau_i c_i \\ \vdots \\ \tau_i c_i \end{bmatrix}. \quad (10)$$

Simulation Descriptions

Simulations are conducted in MATLAB R2022b, where integrals in the predictor-feedback controller (5) are computed using the trapezoidal rule. Moreover, third-order models for vehicles' dynamics, as described in (1)–(3), are implemented using the Euler method. Additionally, a fixed time step of $T_s = 0.01$ is selected to align with realistic sampling times of control execution and measure-

ments, see, e.g., [2].

Table 1: Parameters used for the simulation code in [1, Sect. IV].

Vehicle No. \ Parameters	τ_i	h_i	$D_{c,i}$	m_i
0	0.3	—	0.03	—
1	0.3 s	0.4 s	0.09	1
2	0.25 s	0.4 s	0.12	2
3	0.25 s	0.5 s	0.14	3
4	0.2 s	0.5 s	0.09	3
5	0.25 s	0.3 s	0.18	3
6	0.3 s	0.25 s	0.1	3
7	0.25 s	0.25 s	0.12	3
8	0.25 s	0.5 s	0.14	3
9	0.3 s	0.3 s	—	3

Examples

The simulation results for [1, Sect. IV] can be reproduced by running the script `n2system.m`.

References

- [1] A. Samii, D. Angelopoulos and N. Bekiaris-Liberis, “Predictor-feedback CACC for vehicular platoons with actuation and communication delays based on a multiple-predecessor-following CTH nominal strategy,” *arXiv:2604.05667*, 2026.
- [2] A. Samii, R. de Haan, and N. Bekiaris-Liberis, “Experimental implementation and validation of predictor-based CACC for vehicular platoons with distinct actuation delays,” *IEEE Conference on Decision and Control*, Rio de Janeiro, 2025.